

DevOps / Linux Practical Assignment

Task 1: Secure Multi-User Application Server

Scenario: You've been given a fresh Linux server. Prepare it for a team of developers to deploy and manage an application securely.

- Set up system (hostname, updates, essential tools)
- Create users: dev1, dev2, ops1 and groups: developers, devops
- Configure access: developers can deploy apps; ops has full control (sudo)
- Deploy a simple web app (Nginx/Apache or Docker)
- Secure /var/www/app so only developers can modify
- Grant ops1 elevated privileges
- Disable root SSH login and harden SSH configuration
- Create a custom background service for the app with auto-restart
- Open only required ports (SSH, HTTP) and verify access
- Write a deployment script to update app, restart service, and log actions

Task 2: System Monitoring & Audit Automation

Scenario: Monitor a production server and ensure system health, logging, and alerts are in place.

- Create script system_monitor.sh
- Display CPU, memory, disk usage, top processes, and active users
- Log warning if disk usage exceeds 80%
- Log processes consuming high CPU
- Write output to /var/log/system_monitor.log
- Schedule execution every 5 minutes via cron
- Restrict script execution to authorized users
- Secure log files
- Store logs in mounted directory /monitoring-data with persistence

Task 3: Backup & Recovery with Network Validation

Scenario: Design a backup system for critical application data and ensure it can be restored.

- Create directory /data/app and simulate application data
- Write backup script to compress data and store in /backup with timestamps
- Auto-delete backups older than 7 days
- Log all backup activity
- Schedule daily backup via cron
- Transfer backups using scp/rsync (remote or localhost simulation)
- Validate connectivity before backup
- Restrict access to backup files
- Restore backup to a new directory and verify integrity

Submission Requirements

- Commands and scripts used

- Screenshots or terminal logs
- Short explanation of implementation decisions